

# Project FORESIGHT

NorthBay Living — Demand, Sales & Inventory Dashboard

**Prepared by:** Sheetal Shedashyale · Zidio Development Internship · Data Analyst Track (Power BI)

## What was the problem?

NorthBay Living needed a Power BI dashboard that their team could use directly, without having to work with a data-science notebook. The main goal was to bring demand forecasting, stockout risk, overstock monitoring, and practical business insights into one easy-to-understand report.

## Data used

The project uses the Online Retail II dataset from UCI/Kaggle. It contains about 1,067,371 rows covering December 2009 to December 2011. The data was cleaned and then narrowed to the top 200 SKUs by total quantity sold, matching the stated catalog size while keeping the complete transaction history for those SKUs.

## How the dashboard was built

- Cleaned the raw sales data in Power Query by removing cancellations, non-product codes, invalid prices, and blank SKUs.
- Created a **sku\_master** table and assigned product categories using keyword-based rules.
- Created a calendar table and a simulated **inventory\_snapshots** table containing daily on-hand stock, reorder points, and on-order quantities for each SKU.
- Built a 12-page Power BI report covering Executive Overview, Sales Analytics, Product Performance, Category Performance, Inventory, Stockout Risk, Overstock, Promotion Dashboard, Seasonality Dashboard, Forecast, Insights, and Executive Recommendation.

## Important assumptions

- **Unit cost:** Estimated as 60% of each SKU's average selling price because actual cost data was not available.
- **Inventory data:** Inventory snapshots are simulated, not real company inventory records. They are based on starting stock minus cumulative sales and are used to model on-hand and reorder-point patterns.
- **Category assignment:** Categories are assigned using keywords in product descriptions. For example, MUG/CUP maps to Kitchenware and LIGHT/CANDLE maps to Lighting.
- **Overstock rule:** A SKU is considered overstocked when on-hand stock is at least 1.5 times its reorder point. This threshold was calibrated using the dataset's on-hand/reorder ratio distribution.
- **Reorder status:** As of the latest snapshot date, 26 December 2011, the model shows 0 SKUs below their reorder point. This is a result of the simulated inventory model, not a dashboard error.

# What the dashboard delivers

The final report was designed around the original business requirements and keeps the results understandable for non-technical users.

| BUSINESS NEED         | WHAT WAS DELIVERED                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Demand forecast       | Page 10 — Forecast: Power BI's built-in Analytics forecast with a 90-point projection and 95% confidence interval. |
| Stockout risk         | Page 6 — Stockout Risk: Reorder Now SKU Count card, filtered product table, and an explanatory note.               |
| Overstock monitoring  | Page 7 — Overstock: 73 SKUs flagged as overstocked (36.5%) together with a filtered table.                         |
| Easy-to-use dashboard | All 12 pages, including plain-language Insights on Page 11 and Executive Recommendation on Page 12.                |

## A few key takeaways

- The dashboard brings sales, product, category, inventory, promotion, seasonality, and forecast views into one report.
- The forecast uses Power BI's built-in forecasting capability rather than presenting a separate data-science model.
- The stockout analysis currently shows no SKU below its reorder point at the latest simulated snapshot date.
- The overstock analysis identifies 73 SKUs, or 36.5% of the 200 tracked SKUs, as overstocked under the project's defined threshold.
- The final Insights and Executive Recommendation pages translate the analysis into plain-language information for business users.

## Project note

Because some operational fields were not available in the original retail dataset, inventory and unit-cost values are modeled using the assumptions described above. These assumptions are clearly documented so that the dashboard's results can be understood in the right context.